

Meeting Summaries of March 5

Date: March 5

Time: 18:00 - 19:10 PM(PDT)

Attendees: Prof. Jan, Dr. Javed, Dr. Yue, Jian and Feng

Topic: A **brief view** on the details behind some computation steps; some **suggestions** about explaining research work, and a reminder of **starting writing work now**.

Summary of Discussion

- Feng planed to give an **overview and some details** of the computation steps, and stated that the current computation work is open and updates and intensions can be made without distorting the overall framework.
- However the presentation was **unclear** with several conceptual jumps, skipped the intermediate **reasons behind specific operations**, and only the concrete mathematical operations were shown and **not** connected well to the ecological relevance.
- Nevertheless, Feng **tried to** outline several steps and aspects of the computation, like the format of the taxa data, the subjective merging of clusters and the potential issues behind these operations, but they seemed not well organized at all.
- Regarding the plan for next steps, Feng asked Prof. Jan the **robustness testing method** for the clustering analysis, and proposed that there are **two primary work** to apply and clarified that the mentioned issues are trivial at this moment. Specifically, the clustering testing method and two primary work are:
 1. In the book *Numerical Ecology with R*, the chapter **4.12** provides a brief guidance on constrained (supervised) clustering methods, it focuses on **predictive objective** with **cross-validation** approach to help determine **the number of groups** with the highest predictive accuracy.
 2. The first primary work is to correct the **ZCI construction work**, which **conflated two operations**, Bray-Curtis dissimilarity measure and NMDS ordination appraoch. Their roles are not explained and used clearly and it produced misleading results with interpretations.
 3. The second is to integrate the **Threshold Indicator of Taxa Analysis (TITAN)** method, which may provide comparisons to the detected thresholds and supplements the ecological relevance.
- Jian suggested that Feng should use **plainer language** to explain the modeling part, and a clearer connection to the ecological relevance. A **one-to-one meeting** can be arranged to discuss the details behind the original analysis results.

- Dr. Javed suggested to supply more (maybe one or two) **reasons** behind each mathematical operation, like why and how a specific transformation step is needed. **Specifically**, briefly outlining the mathematical operations and commenting how it helps to achieve a better statistical work and a clearer ecological relevance, then what specific data inputs on level of granularity are passed into such operation.
- Prof. Jan reminded Feng to make and share the **plan of weekly goals** for the thesis writing and defence, it is important to have a controlled and visible dashboard of the timeline.

Decisions Made

- Jian will arrange a one-to-one **meeting** with Feng to further understand the work and give advice on the writing and presenting work.
- Feng will supply **reasons** for operations and its details that are worth explaining, **balancing** the time, clearness and depth in both the writing and the presentation.
- Feng will finish the **plan of weekly goals** for the thesis writing and defence, and share it.
- Feng is going to **finish** the two primary work and **update** the clustering analysis of the book. Meanwhile, Feng found the next chapter **4.15** of the book provides a more recent method which may be a good **alternative** to try.
- Feng is **starting thesis writing** work now, there should be some formal thesis writing to support the weekly meetings and the weekly goals from now on, so that the writing can be done in expected timeline.

Follow-Up

Temporary meeting time: March 12, 18:00 - 18:40 PM (Vancouver), which corresponds to 19:00 - 19:40 PM (Calgary) and 20:00 - 20:40 PM (Winnipeg)

Supportive materials

- A meeting recording with subtitles has been restored on local drive.